

Buckert experiment results

The aim of the experiment was to find out if exposure has a significant effect on the measurement of the Volumetric Water Content with TOMST TMS4 sensors.

Therefore, three buckets were filled with sand and tap water was added till saturation was reached. Afterwards, sensors were placed as described below. The buckets were left to dry under stable conditions from 05.11.2025 (3pm) to 09.12.2025 (11.30 am). The period used for calculations was 06.11.2025 to 08.12.2025. That was to give the sensors time to adjust to the setting.

The experiment contains 3 buckets in total. Bucket two and three contained 4 sensor each. Two of them not exposed, two of them 6 cm exposed. (During data collection it turned out, that the two sensors in bucket three were actually more like 5 cm exposed) Bucket two was placed in room temperature conditions, Bucket three in the fridge to imitate more antarctica like temperatures.

Bucket one was larger than the equally sized buckets two and three. Therefore it contained 6 sensors, 0, 3, and 6 centimetres exposed and two sensors for each exposure step.

Bucket one



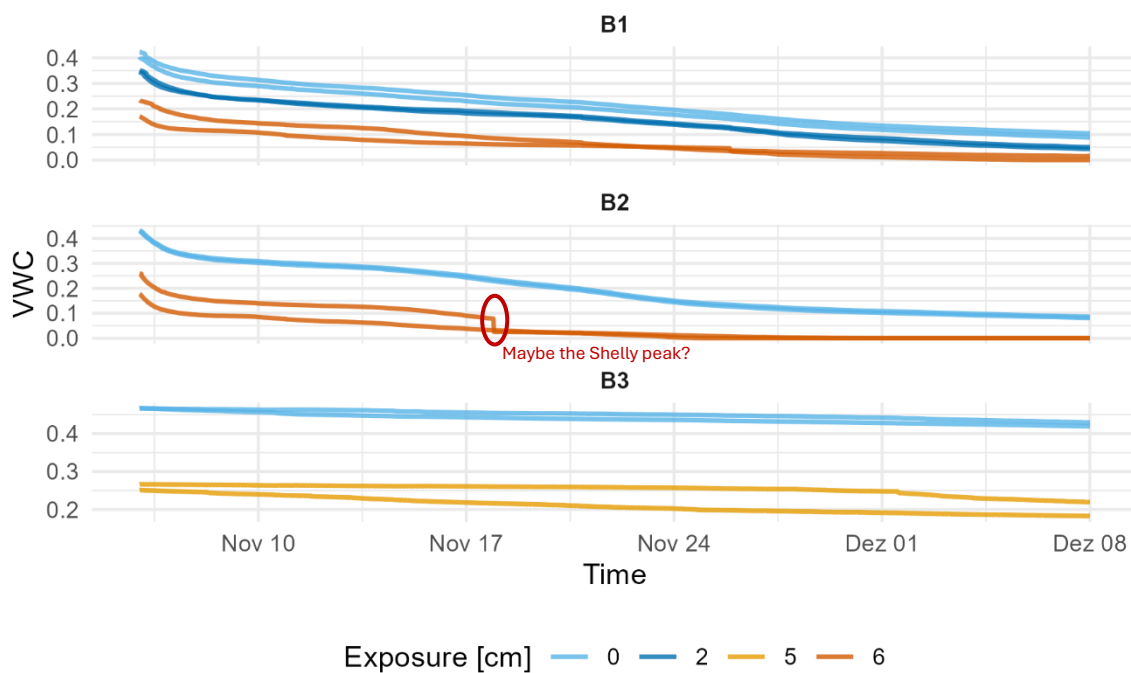
Bucket two



Bucket 3; Sensors were 5 cm exposed not 6 as originally planned.



WVC over time by bucket, logger and exposure



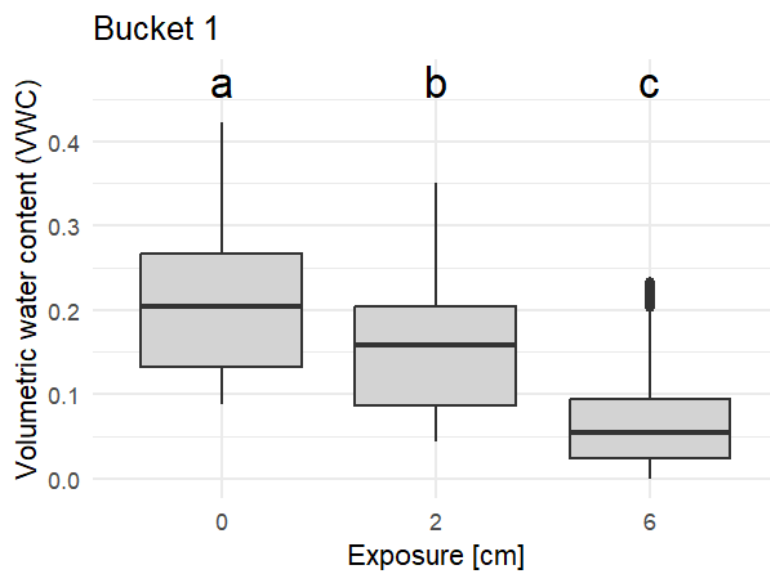
For all Buckets; data: VWC by exposure [cm], the marked “Shelly Peak” is a result of an unintended contact which most likely had led to a slightly different logger position.

Bucket 1

Kruskal-Wallis chi-squared = 8607, df = 2, p-value < 2.2e-16

Dunn (1964) Kruskal-Wallis multiple comparison
p-values adjusted with the Holm method.

	Comparison	Z	P.unadj	P.adj
1	0 - 2	30.07577	1.005411e-198	1.005411e-198
2	0 - 6	91.04340	0.000000e+00	0.000000e+00
3	2 - 6	60.96764	0.000000e+00	0.000000e+00



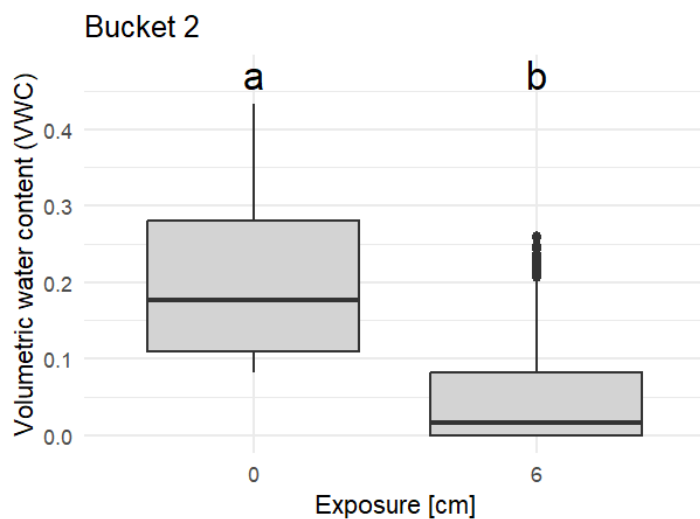
Bucket 2

Wilcoxon rank sum test with continuity correction

data: value by exposure_cm

W = 34914109, p-value < 2.2e-16

alternative hypothesis: true location shift is not equal to 0



Bucket 1 and 2 combined (not fridge)

max(bucket_2_VWC\$value) = 0.4334356

max(bucket_1_VWC\$value) = 0.4230493

➔ Comparable start conditions

Kruskal-Wallis rank sum test

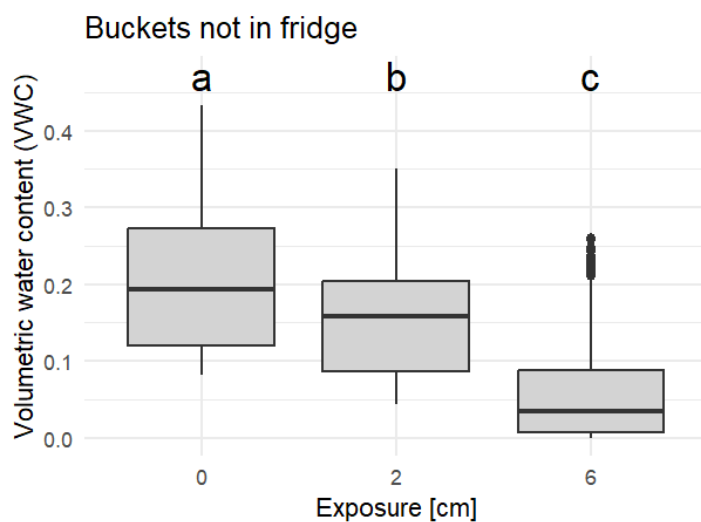
data: value by exposure_cm

Kruskal-Wallis chi-squared = 15678, df = 2, p-value < 2.2e-16

Dunn (1964) Kruskal-Wallis multiple comparison

p-values adjusted with the Holm method.

	Comparison	Z	P.unadj	P.adj
1	0 - 2	26.92673	1.068463e-159	1.068463e-159
2	0 - 6	122.62189	0.000000e+00	0.000000e+00
3	2 - 6	73.19362	0.000000e+00	0.000000e+00

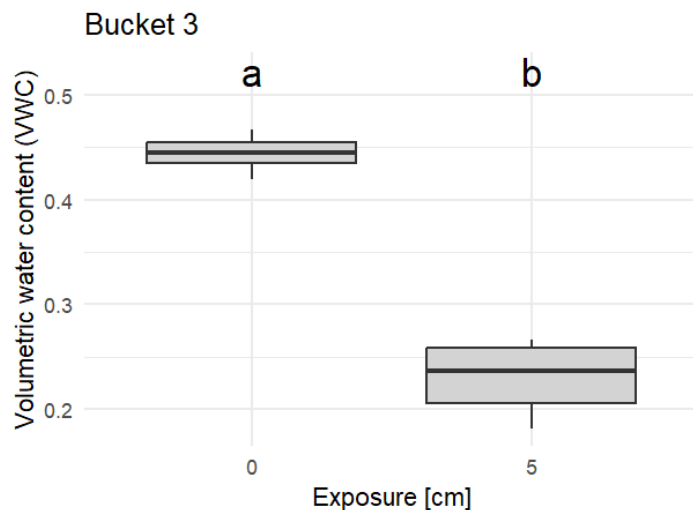


Bucket 3 (fridge)

Wilcoxon rank sum test with continuity correction

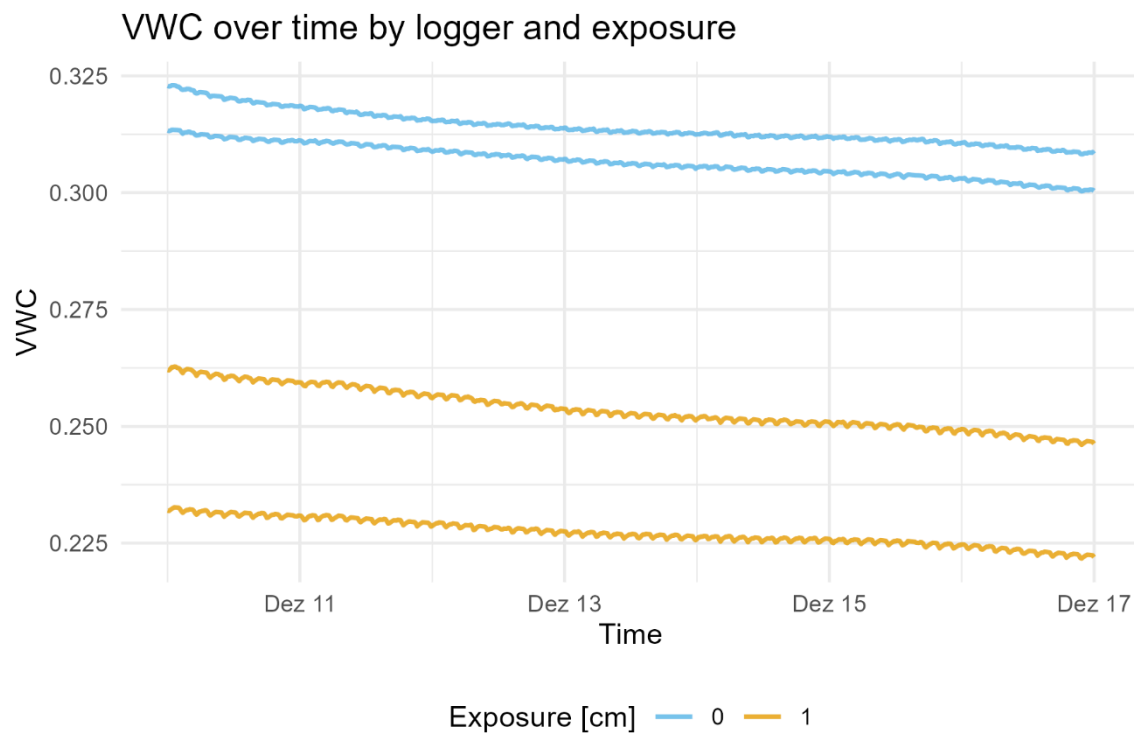
data: value by exposure_cm

W = 37773316, p-value < 2.2e-16



Additional bucket experiment

4 sensors were placed in the bucket from the fridge for another week. Two of them were not exposed, the other two were 1cm exposed. The fridge was chosen to investigate their behaviour in more or less stable (slowly drying) wet conditions. Please notice that one of the approximately 1cm exposed loggers was a bit more exposed than the other one. The sensitivity of the loggers is 1%.



```
wilcox.test(value ~ exposure_cm, data = add_bucket_VWC)
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Wilcoxon rank sum test with continuity correction

data: value by exposure_cm

W = 1811716, p-value < 2.2e-16

alternative hypothesis: true location shift is not equal to 0

Interpretation:

Also the 1 cm exposure situation is significantly influencing the VWC measurement. Please note that the Volumetric Water content measurements were rounded to the logger sensitivity previous to the Wilcoxon test.

In summary, all exposure steps did make a difference. The impact is a reduction of the measured Volumetric water content with increasing exposure. This is making sense because the physically measured signal will be reduced as a result of the exposure.

I further assume that if we have no significant difference between two loggers placed in the same conditions, then we can assume that they are either not exposed or equally exposed.